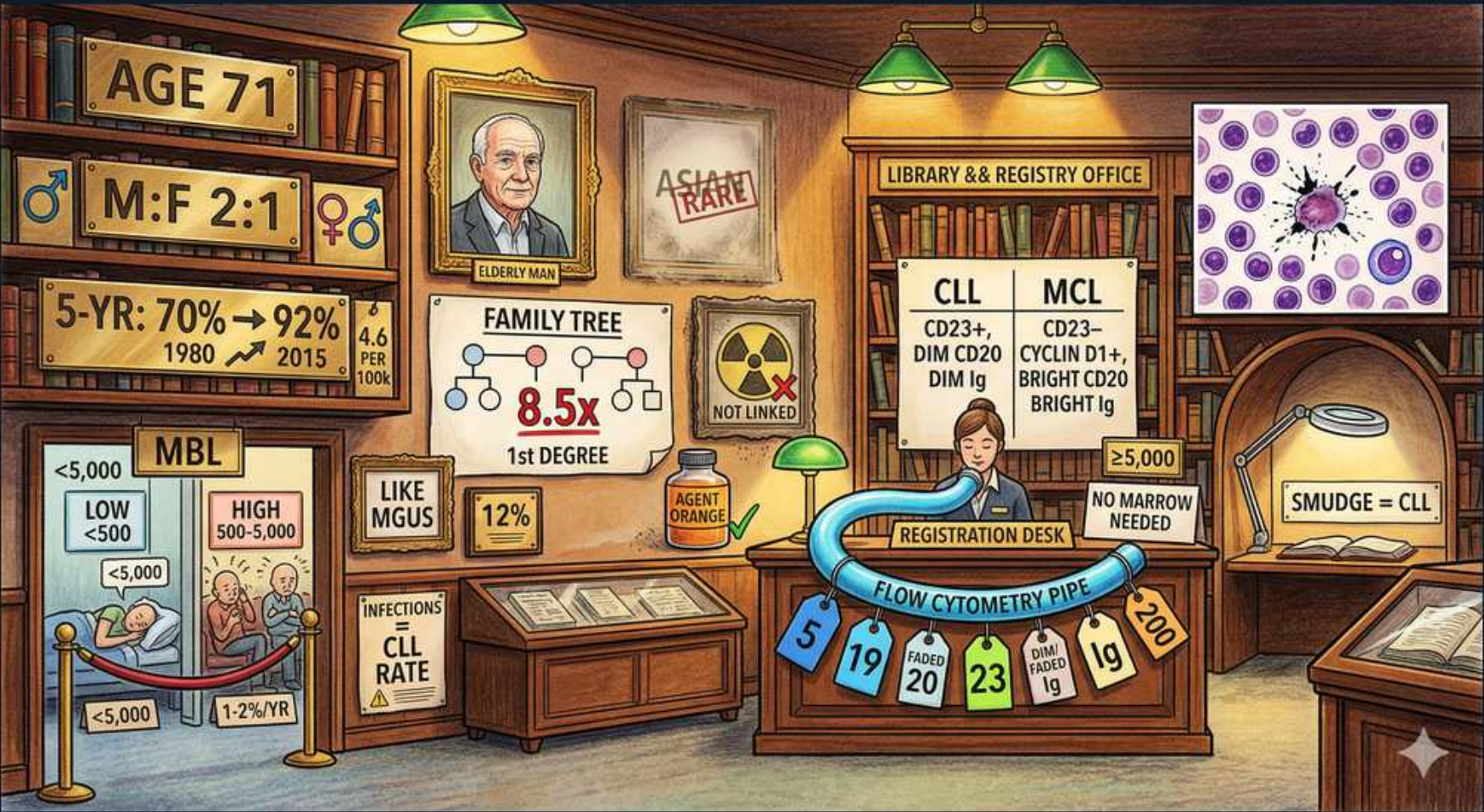


CLL — Epidemiology, Risk Factors & Diagnosis



1. Age 71 / Median

WHAT YOU SEE

'AGE 71' framed sign, elderly man portrait

WHAT IT ENCODES

CLL is a disease of older adults — median age at diagnosis ~70-72. It is the most common adult leukemia in the Western world. Rare under age 40.

BOARD TRAP

A young patient (<40) with lymphocytosis should raise suspicion for a different entity (mantle cell, reactive) — CLL would be atypical.

2. Male Predominance 2:1

WHAT YOU SEE

'M:F 2:1' sign with gender symbols

WHAT IT ENCODES

CLL has a male-to-female ratio ~1.5-2:1. More common in males and in people of European descent; rare in East Asians (genetic, not environmental).

BOARD TRAP

The low incidence in Asians is genetic — it persists after migration, unlike environmentally driven cancers.

3. 5-Yr Survival 70%→92%

WHAT YOU SEE

'5-YR 70%→92%' 1980 to 2015 trend arrow

WHAT IT ENCODES

5-year survival improved dramatically (~70% to >90%) with novel targeted agents (BTK inhibitors, BCL2 inhibitors) replacing chemo. Many patients die WITH rather than FROM CLL.

BOARD TRAP

Improved survival reflects targeted therapy era — don't attribute it to chemotherapy intensification, which actually fell out of favor.

4. Family Risk 8.5x

WHAT YOU SEE

Family tree, '8.5x' bold, 1st DEGREE label

WHAT IT ENCODES

CLL has the strongest familial clustering of any leukemia — ~8.5x increased risk in first-degree relatives. No single causative gene; polygenic inheritance.

BOARD TRAP

Despite strong heritability, routine screening of relatives is NOT recommended — most are asymptomatic and management wouldn't change.

5. Not Linked to Radiation

WHAT YOU SEE

Radiation trefoil with 'NOT LINKED' sign

WHAT IT ENCODES

Unlike AML/CML, CLL is NOT associated with ionizing radiation. Atomic bomb survivors had no increased CLL. Agent Orange (herbicides) IS a recognized association.

BOARD TRAP

Classic board trap: radiation causes most leukemias but NOT CLL. Agent Orange/pesticide exposure is the recognized environmental link.

6. Agent Orange Link

WHAT YOU SEE

'AGENT ORANGE' tag near family/exposure area

WHAT IT ENCODES

Agent Orange and other herbicide/pesticide exposures are recognized CLL risk factors (VA presumptive condition). Farming exposure also implicated.

BOARD TRAP

Don't pick radiation as the CLL exposure — herbicides (Agent Orange), not ionizing radiation, are the recognized association.

7. MBL Precursor

WHAT YOU SEE

'MBL' panel, <5,000, low vs high, 1-2%/yr

WHAT IT ENCODES

Monoclonal B-cell lymphocytosis (MBL) = clonal B-cells <5,000/ μ L without other CLL features. It is the obligate precursor to CLL. High-count MBL progresses to CLL at ~1-2% per year.

BOARD TRAP

MBL is defined by <5 $\times$ 10³/L clonal B-cells AND no cytopenias/organomegaly — cross 5,000 or develop symptoms and it becomes CLL.

8. CLL $\geq$ 5,000 Threshold

WHAT YOU SEE

' $\geq$ 5,000' sign, registration desk, flow cytometry pipe

WHAT IT ENCODES

Diagnosis of CLL requires $\geq$ 5 $\times$ 10³/L (5,000/ μ L) clonal B-lymphocytes in peripheral blood for $\geq$ 3 months, confirmed by flow cytometry. Below this with no other features = MBL.

BOARD TRAP

The 5,000 threshold is for B-cells specifically, not total lymphocytes. If tissue (node) involvement without blood criteria → small lymphocytic lymphoma (SLL), same disease.

9. Flow Immunophenotype (CD5/CD19/CD23)

WHAT YOU SEE

Registry office: CLL CD23+/dim CD20/dim Ig; flow cytometry pipe with markers 5,19,23,20,200

WHAT IT ENCODES

CLL phenotype: CD5+, CD19+, CD23+, CD20 dim, surface Ig dim, FMC7-. The aberrant CD5 co-expression (a T-cell marker) on clonal B-cells is the hallmark.

BOARD TRAP

CD5+ is shared with mantle cell lymphoma — distinguish by CD23 (CLL+ / MCL-), cyclin D1 (MCL+), and t(11;14). Dim CD20/Ig favors CLL.

10. CLL vs MCL Discriminator

WHAT YOU SEE

Side-by-side CLL vs MCL columns: CD23, CD20, Ig, cyclin D1

WHAT IT ENCODES

MCL is also CD5+ but CD23-negative, CD20 BRIGHT, surface Ig bright, cyclin D1+ with t(11;14). CLL is CD23+, dim CD20/Ig. Getting this wrong changes prognosis dramatically (MCL is aggressive).

BOARD TRAP

Both CD5+ — the trap is stopping at CD5. CD23 and cyclin D1/t(11;14) are the discriminators; mislabeling MCL as CLL undertreats an aggressive lymphoma.

11. Smudge Cells

WHAT YOU SEE

'SMUDGE = CLL' sign, purple smeared cells in microscope inset

WHAT IT ENCODES

Smudge (basket) cells on the peripheral smear are fragile CLL lymphocytes crushed during slide prep — a classic CLL clue. Abundant smudge cells correlate with CLL diagnosis.

BOARD TRAP

Smudge cells are an artifact of fragile cells — adding albumin to the smear reduces them. They support CLL but flow cytometry confirms.

12. No Marrow Biopsy Needed

WHAT YOU SEE

'NO MARROW NEEDED' sign

WHAT IT ENCODES

CLL is diagnosed on peripheral blood flow cytometry alone — bone marrow biopsy is NOT required for diagnosis. Marrow may be done to evaluate cytopenias of unclear cause before therapy.

BOARD TRAP

Don't order a marrow biopsy to diagnose CLL — blood flow suffices. Biopsy is reserved for clarifying whether cytopenias are marrow infiltration vs autoimmune/treatment.